

**SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022****Mathematics****MAT 6B 09—REAL ANALYSIS****(2014 to 2018 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 30****Maximum : 30 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 30.
2. The candidate should check that the question paper supplied to him/her contains all the 30 questions in serial order.
3. Each question is provided with choices (A), (B) and (C) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

## MAT 6B 09—REAL ANALYSIS

(Multiple Choice Questions for SDE Candidates)

1. Let  $A$  be the open interval  $(-1, 1)$ . Then, the cluster points of  $A$  are :

- (A) 1 and  $-1$  only.
- (B) All the points of  $(-1, 1)$ .
- (C) All the points of  $[-1, 1]$ .

2. Which of the following have no cluster points ?

- (A)  $(3, 4]$ .
- (B)  $[2, 3]$ .
- (C)  $\{2, 3, 4\}$ .

3. Which among these functions are continuous on  $A = \{x \in \mathbb{R}, x \geq 0\}$  ?

- (A)  $f(x) = \sin x$ .
- (B)  $f(x) = \sin\left(\frac{1}{x}\right)$ .
- (C)  $f(x) = \frac{1}{x}$ .

4. Let  $f(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$  Then  $\lim_{x \rightarrow 0} f(x)$  is :

- (A) 1.
- (B)  $-1$ .
- (C) None of these.

5. Given  $a, b$  satisfy  $0 < a < 1, b > 1$ . Then which of the following sequences is convergent :

- (A)  $\frac{ab^n}{2^n}$ .
- (B)  $\frac{2^{3n}}{3^{2n}}$ .
- (C)  $\frac{n}{b^n}$ .

6. Function  $f(x) = \frac{1}{2x}$  defined on  $A = (0, \infty)$  is an example for :
- (A) A continuous function that is not bounded.
  - (B) A bounded function that is continuous.
  - (C) An unbounded function that is not continuous.
7. Let  $f : A \rightarrow \mathbb{R}$  and let  $L$  be a real number such that for every  $\epsilon > 0$ , there exist a  $\delta(\epsilon) > 0$  such that if  $x \in A$  and  $0 < |x - c| < \delta(\epsilon)$ , then  $|f(x) - L| < \epsilon$ . Then  $L$  is a \_\_\_\_\_ of  $f$  at  $c$ .
- (A) Derivative.
  - (B) Limit.
  - (C) None of these.
8. When the graph of a function  $f$  in  $\mathbb{R}$  has a break such that the function value and the limit (does not exist) aren't the same, then  $f$  is said to have a \_\_\_\_\_ discontinuity at that point.
- (A) Removable.
  - (B) Jump.
  - (C) None of these.
9. When the graph of a function  $f$  in  $\mathbb{R}$  has a hole in the graph such that the function value and the limit aren't the same, then  $f$  has a \_\_\_\_\_ discontinuity at that point.
- (A) Removable.
  - (B) Jump.
  - (C) None of these.
10. Let  $A \subseteq \mathbb{R}$  and let  $f : A \rightarrow \mathbb{R}$ . If there exist a point  $x' \in A$  such that  $f(x') \geq f(x)$ , for all  $x \in A$ , then  $f$  has an \_\_\_\_\_ on  $A$ .
- (A) Absolute maximum.
  - (B) Absolute minimum.
  - (C) None of these.
11. Let  $f : I \rightarrow \mathbb{R}$  be continuous. Then  $f$  has an absolute maximum and absolute minimum on  $I$  if :
- (A)  $I$  is closed.
  - (B)  $I$  is closed and bounded not bounded.
  - (C)  $I$  is bounded but not closed.

12. Consider  $f(x) = 6x^3 - 3x^2 + 2$ . Then which of the following is true ?
- (A)  $f$  has a zero in  $(0, 1)$ .
  - (B)  $f$  has no zero.
  - (C)  $f$  has a zero in  $(-1, 1)$ .
13. Which of the following statements is true ?
- (A) Every continuous function is uniformly continuous.
  - (B) Every uniformly continuous function is continuous.
  - (C) None of these.
14. Suppose  $f$  and  $g$  are in  $R[a, b]$  such that  $f(x) \leq g(x)$  for all  $x \in [a, b]$ , then :
- (A)  $\int_a^b f(x) < \int_a^b g(x)$ .
  - (B)  $\int_a^b f(x) \leq \int_a^b g(x)$ .
  - (C) None of these.
15. Which of the following is true ?
- (A)  $f \in R[a, b] \Rightarrow f$  is bounded.
  - (B)  $f \in R[a, b] \Rightarrow f$  is continuous.
  - (C)  $f \in R[a, b] \Rightarrow f$  is monotone.
16. Let  $I = [2, 9]$  which of the following partitions of  $I$  has the largest norm ?
- (A)  $P_1 = (2, 4, 6, 8, 9)$ .
  - (B)  $P_2 = (2, 6, 9)$ .
  - (C)  $P_3 = (2, 5, 7, 9)$ .
17. Let  $x \in [0, 1]$  and  $f$  be such that  $f(x) = \begin{cases} 1, & x \text{ is rational} \\ 0, & x \text{ is irrational} \end{cases}$ . Then :
- (A)  $f \in R[0, 1]$ .
  - (B)  $f \notin R[0, 1]$ .
  - (C) None of these.

18. Suppose  $f : [a, b] \rightarrow \mathbb{R}$  and  $f(x) = 0$  except for finite number of points  $c_1, c_2, \dots, c_n$  in  $[a, b]$ . Then :
- (A)  $f \notin R[a, b]$ .
  - (B)  $\int_a^b f = 0$ .
  - (C) None of these.
19. Let  $f \in R[a, b]$  and  $c \in \mathbb{R}$ . Define  $g$  on  $[a + c, b + c]$  by  $g(y) = f(y - c)$ . Then which of these holds ?
- (A)  $\int_a^b f = \int_a^b g$ .
  - (B)  $\int_a^b f = \int_{a+c}^{b+c} g$ .
  - (C)  $g \in R[a, b]$ .
20. Which of the following is true ?
- (A) Every convergent sequence is Cauchy.
  - (B) Every Cauchy sequence is convergent.
  - (C) Both (A) and (B).
21. Let  $f : [a, b] \rightarrow \mathbb{R}$ . Then which of the following is true ?
- (A)  $f$  is continuous on  $[a, b] \Rightarrow f \in R[a, b]$ .
  - (B)  $f \in R[a, b] \Rightarrow f$  is continuous on  $[a, b]$ .
  - (C)  $f \in R[a, b] \Rightarrow f$  is uniformly continuous on  $[a, b]$ .
22. Let  $f : [a, b] \rightarrow \mathbb{R}$ . Then which of the following is true ?
- (A)  $f$  is monotone on  $[a, b] \Rightarrow f \in R[a, b]$ .
  - (B)  $f \in R[a, b] \Rightarrow f$  is monotone on  $[a, b]$ .
  - (C) None of these.

23. Let  $f$  be continuous on  $[a, b]$  and  $f(x) \geq 0$  for all  $x \in [a, b]$ . If  $\int_a^b f = 0$ , then which of the following is true ?

- (A)  $f(x) = 0$  for all  $x \in [a, b]$ .
- (B)  $f(x) = 0$  for all but finitely many points of  $[a, b]$ .
- (C) None of these.

24. If  $f$  is bounded on  $[a, b]$  and if  $f$  restricted to  $[c, b]$ ,  $c \in (a, b)$  is Riemann integrable, then which of the following does not hold true ?

- (A)  $f \in R[a, b]$ .
- (B)  $f \notin R[a, b]$ .
- (C)  $\lim_{c \rightarrow a^+} \int_c^b f = \int_a^b f$ .

25. Define ' $g$ ' in  $[0, 1]$  as  $g(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$  Then which of these is true ?

- (A)  $g \in R[0, 1]$ .
- (B)  $g \notin R[0, 1]$ .
- (C) None of these.

26. If ' $f$ ' is continuous on  $[-1, 1]$  then  $\int_0^{\pi/2} f(\cos x) \cdot dx$  is not equal to which of the following ?

- (A)  $\int_0^{\pi/2} f(\sin x) \cdot dx$ .
- (B)  $\frac{1}{2} \int_0^{\pi} f(\sin x) \cdot dx$ .
- (C)  $\int_0^{\pi} f(\sin x) \cdot dx$ .

27. If  $f$  is continuous at every point of  $[a, b]$  and  $F$  is any antiderivative of  $f$  on  $[a, b]$ , then  $\int_a^b f(x) \cdot dx =$

(A)  $(b - a)(F(b) - F(a)).$

(B)  $F(b) - F(a).$

(C) None of these.

28. If  $g(x) = \begin{cases} x, & |x| \geq 1 \\ -x, & |x| < 1 \end{cases}$  and if  $G(x) = \frac{1}{2}|x^2 - 1|$ , then  $\int_{-2}^3 g(x) dx =$

(A)  $\frac{1}{2}.$

(B)  $\frac{5}{2}.$

(C)  $\frac{7}{2}.$

29. Let  $B(x) = \begin{cases} -\frac{1}{2}x^2, & x < 0 \\ \frac{1}{2}x^2, & x \geq 0. \end{cases}$  Then  $\int_a^b |x| dx =$

(A)  $(b - a)[B(b) - B(a)].$

(B)  $B(b) - B(a).$

(C) None of these.

30. If  $f : [0, 1] \rightarrow \mathbb{R}$  is continuous and  $\int_0^x f = \int_x^1 f$  for all  $x \in [0, 1]$ , then :

(A)  $f(x) = 0, \forall x \in [0, 1].$

(B)  $f \in R[0, 1].$

(C) All the above.

**SIXTH SEMESTER (CUCBCSS-UG) DEGREE EXAMINATION, MARCH 2022**

Mathematics

MAT 6B 09—REAL ANALYSIS

(2014 to 2018 Admissions)

Time : Three Hours

Maximum : 120 Marks

**Part A**

*Answer all questions.  
Each question carries 1 mark.*

1. Define uniform continuity of a function.
2. State Weierstrass approximation theorem.
3. Find  $\|P\|$  if  $P = \{0, 2, 3, 4\}$  in a partition of  $[0, 4]$ .
4. Give an example for a function which is not Riemann integrable.
5. Define step function.
6. State Lebesgue integrability criterion.
7. Define uniform convergence of a series of functions.
8.  $\lim_{n \rightarrow \infty} \frac{x^2 + nx}{n}$ .
9. Write an example for an absolutely convergent improper integral.
10. Cauchy principal value of  $\int_{-\infty}^{\infty} x dx =$ .
11. Define Beta function.
12. Fill in the blanks :  $\Gamma(3) =$  \_\_\_\_\_.

(12 × 1 = 12 marks)

**Part B**

*Answer any ten questions.  
Each question carries 4 marks.*

13. Let  $I = [a, b]$  be a closed bounded interval and  $f : I \rightarrow \mathbb{R}$  be continuous on  $I$ . If  $k \in \mathbb{R}$  is any number satisfying  $\inf f(I) \leq k \leq \sup f(I)$  then prove that there exists a number  $c \in I$  such that  $f(c) = k$ .
14. State and prove Preservation of Intervals theorem.

**Turn over**

15. Show by an example that every uniformly continuous function need not be a Lipschitz function.
16. If  $\phi: [a, b] \rightarrow \mathbb{R}$  is a step function, prove that  $\phi \in \mathcal{R}[a, b]$ .
17. Suppose that  $f, g \in \mathcal{R}[a, b]$ . Prove that  $fg \in \mathcal{R}[a, b]$ .
18. State the substitution theorem of Riemann integration. Use it to evaluate  $\int_0^4 \frac{\sin \sqrt{t}}{\sqrt{t}} dt$ .
19. Let  $(f_n)$  be a sequence of bounded functions of  $A \subseteq \mathbb{R}$ . Suppose that  $\|f_n - f\|_A \rightarrow 0$ . Then prove that  $(f_n)$  converges uniformly on  $A$  to  $f$ .
20. If  $f_n$  is continuous of  $D \subseteq \mathbb{R}$  to  $\mathbb{R}$  for each  $n \in \mathbb{N}$  and if  $\sum f_n$  converges to  $f$  uniformly on  $D$ , then prove that  $f$  is continuous on  $D$ .
21. State and prove Weierstrass M-Test for a series of functions.
22. Discuss the uniform convergence of  $\sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$ .
23. Test the convergence of  $\int_0^{\infty} \frac{1}{x^2} dx$ .
24. Show that  $\Gamma(n+1) = n!$  when  $n$  is a positive integer.
25. Show that  $\beta(m, n) = \beta(n, m)$ .
26. Evaluate  $\int_0^1 x^m \left( \log \frac{1}{x} \right)^n dx$ .

(10 × 4 = 40 marks)

**Part C**

*Answer any six questions.  
Each question carries 7 marks.*

27. Let  $I = [a, b]$  be a closed bounded interval and let  $f: I \rightarrow \mathbb{R}$  be continuous on  $I$ . Then prove that  $f$  is bounded on  $I$ .
28. State and prove Uniform Continuity Theorem.
29. State and prove Continuous Extension Theorem.
30. If  $f: [a, b] \rightarrow \mathbb{R}$  is monotone on  $[a, b]$  then prove that  $f \in \mathcal{R}[a, b]$ .
31. Discuss the convergence of the sequence  $(f_n(x))$  where  $f_n(x) = \frac{x^n}{x^n + 1}$ ,  $x \in [0, 2]$ .
32. State and prove Taylor's Theorem with the Remainder.

33. Let  $f \in \mathcal{R}[a, b]$  and let  $f$  be continuous at a point  $c \in [a, b]$ . Prove that the indefinite integral

$$F(z) = \int_a^z f \text{ for } z \in [a, b] \text{ is differentiable at } c \text{ and } F'(c) = f(c).$$

34. Show that  $\Gamma\left(\frac{p}{2}\right)\Gamma\left(\frac{p+1}{2}\right) = \frac{\sqrt{\pi}}{2^{p-1}}\Gamma(p)$ .

35. Evaluate the integral  $\int_0^1 x^2(1 - \sqrt{x})dx$ .

(6 × 7 = 42 marks)

### Part D

*Answer any two questions.*

*Each question carries 13 marks.*

36. (a) State and prove Maximum Minimum Theorem.

(b) Test the uniform continuity of  $f(x) = \sqrt{x}$  on  $[0, 2]$ .

37. (a) Let  $f : [a, b] \rightarrow \mathbb{R}$  and  $c \in (a, b)$ . Prove that  $f \in \mathcal{R}[a, b]$  if and only if its restriction to  $[a, c]$  and  $[c, b]$  are both Riemann integrable. In this case show that  $\int_a^b f = \int_a^c f + \int_c^b f$ .

(b) If  $f \in \mathcal{R}[a, b]$  and if  $[c, d] \subseteq [a, b]$  then prove that the restriction of  $f$  to  $[c, d]$  is in  $\mathcal{R}[c, d]$ .

38. (a) Prove that  $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}, \forall m, n > 0$ .

(b) Find the value of  $\Gamma\left(\frac{1}{2}\right)$ .

(2 × 13 = 26 marks)